

Erwin POUSSI

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Building autonomous systems that learn to act — from foundation model training and sim-to-real robot deployment to LLM-driven scientific agents.

EDUCATION

Stanford University, Palo Alto (CA) 2025 – 2027

- MS in Aeronautics & Astronautics GPA: **4.1/4.0**
- Relevant coursework: *Principles of Robot Autonomy (AA274A/CS237A)*, *Introduction to Robotics (CS223A)*, *Decision Making under Uncertainty (AA228/CS238)*, *Optimal and Learning-Based Control (AA203)*, *Deep Learning & Reinforcement Learning (CS224R)*.

Ecole Polytechnique, Palaiseau (FRA) 2020 – 2025

- Master's level in Mechanical Engineering, concentration in Autonomy, minor in Entrepreneurship. GPA: **3.96/4.0**
- Relevant coursework: *Advanced Machine Learning for Autonomous Agents*, *Optimization and Control*, *Statistics*.

WORK EXPERIENCE

Stanford Multi-Robot Systems Lab Palo Alto, CA Jan. 2025 – Present

- VLA Training — Imitation Learning & Sim-to-Real** (Jan.–Apr. 2025, Prof. Schwager): Trained VLA policies for autonomous drone navigation via behavioral cloning and DAgger. ([project page](#))
 - Reached **90% navigation success** (up from 52%) using RRT* planning, MPC expert demonstrations, and CLIPSeg grounding to 3D Gaussian Splats.
- GAIT — Agentic Scene Understanding for Robot Navigation** (Jun.–Sep. 2026, Prof. Schwager): Built a graph-memory architecture and robotic navigation framework for VLA models, leveraging online semantic scene graph construction with interrupt-triggered re-planning. ([project page](#))
- GAIT-Manip — Whole-Body Robot Control for Loco-Manipulation** (Sep.–Dec. 2026, Profs. Pavone & Schwager): Whole-body control of G1 and Spot robots for loco-manipulation, using GAIT scene reasoning. ([project page](#))

Stanford School of Medicine, Qiu Lab (Prof. Qiu) Palo Alto, CA Oct. 2025 – Present

- ML Researcher — Pantheon-CLI:** Multi-agent LLM framework for autonomous scientific discovery with tool-augmented inference for genomic analysis. ([project page](#))
 - Fine-tuned Qwen3 LLMs via SFT + RLOO for a self-evolving agent backbone; built actor-critic RL for gene panel selection in scRNA-seq.
- Virtual Embryo Challenge — 3D Convolutional Models for Embryogenesis** (Jun.–Sep. 2026): Training foundational 3D models to reproduce embryogenesis at both cell-level gene expression and morphogenesis; co-building the [Virtual Embryo Challenge](#). ([project page](#))

Inspirit AI San Francisco, CA Jul. 2026 – Aug. 2026

- AI Instructor:** Taught ML, neural networks, and AI for social good to scholars; designed hands-on sessions and guided real-world AI projects culminating in deployed applications.

Stevens Institute of Technology, RRG (Prof. Rabinovitch) Hoboken, NJ Apr. 2025 – Aug. 2025

- Research Intern — ML-Based Aerodynamic Modeling:** Multi-fidelity ML framework (random forests + genetic algorithms) for parachute permeability under rarefied conditions; **<1% mean error**. ([project page](#))

SELECTED RESEARCH

Reinforcement Learning for Algorithmic Trading Jan. 2025 – Apr. 2025

- Built a custom high-frequency market simulator and trained agents with PPO, Actor-Critic, and DQL, focusing on reward-free exploration, robustness to distribution shifts, and real-time decision-making.

Guidance System for Autonomous Visual-Based Docking — Nyx Spacecraft (CSEP) Sep. 2023 – May 2024

- Undergraduate thesis on vision-based navigation for the Nyx spacecraft (selected by Axiom for ISS support). Supervised by The Exploration Company with Polytechnique Space Center. ([project page](#))

TECHNICAL SKILLS

Programming & Frameworks: Python, PyTorch, CUDA, ROS2, MuJoCo, Hugging Face / Transformers, ACADOS, GSplat, GTSAM, CLIPSeg, Habitat-Sim, W&B

Methods: Behavioral cloning, DAgger, PPO, Actor-Critic, DQL, RRT*, MPC, SFT, RLOO, RLVR, LLM fine-tuning, imitation learning, sim-to-real transfer, motion planning, trajectory optimization

Languages: French (native), English (TOEFL 109), Mooré (fluent), German (basic)

PUBLICATIONS

Xu, W., Poussi, E., et al. *PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery*. Preprint, 2026. DOI: [10.64898/2026.02.26.707870](#)

EXTRACURRICULAR ACTIVITIES

Volunteering: Stanford BGSA Fundraising Chair (2025–Present) · Cheer Up cancer nonprofit (2023–2025) · XProjects consulting PM (2024–2025) · Math/physics tutor (2022–2023)

Sports: Handball (France's intercollegiate championship), Track & Field, Weightlifting **Arts:** Drawing, Painting, Dancing

Awards: French Gov. Excellence Scholarship · 2nd, 2021 Youth Olympiads (Legion of Honor) · 1st nationwide, 2020 Burkina Faso A-levels · 2020–18 Excellence Awards